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Second Mid-Term Examination, Odd Semester 2019-20

Program: B-Tech-CSE, **Year:** II **Semester:** III

BCSC 0004: Operating Systems

Time: 1 Hour **Maximum Marks: 15**

Note: Attempt All the Questions

Section – A 3 x 2 = 6 Marks

I. Draw the **precedence graph** for the given code and **rewrite the statements** in precedence graph using parbegin(cobeg) and parend(coend) construct

```
S1;                                  // Statement
Count: = 3;                          // No. of process to be joined
FORK L1;
S2;                                  // Statement
S4;                                  // Statement
FORK L2;
S5;                                  // Statement
GO TO L3;
L2: S6;                              // Statement
    GOTO L3;
L1: S3;                              // Statement
L3: JOIN count;
S7;                                  // Statement
```

II. Consider the following snapshot of a system for Banker's Algorithm having 4 processes P0 to P3 and 3 resources A, B, C.

	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
P0	1	0	1	4	3	1	3	3	0
P1	1	1	2	2	1	4			
P2	1	0	3	1	3	3			
P3	2	0	0	5	4	1			

A. Find out the **Need matrix** for the given above information.

B. Is the system safe? If yes, find out the **safe sequence**.

III. Explain **Deadlock** in brief. If a single resource is allowed to be shared by two processes at same time, justify whether indefinite blocking will be **violated or not**. If yes, name which condition will be violated among all four necessary conditions. Also explain the term Inter Process Communication (**IPC**).

Section – B

3 x 3 = 9 Marks

I. Consider 2 processes P1 and P2 and a **common shared buffer** between them. Assuming the following algorithm, Design a **synchronized code** for P1 and P2 process to achieve mutual exclusion between them.

P1: Checks whether the buffer has at least one empty slot (if not it will wait to get at least one slot to be free and will be blocked till then), if found empty slot, it acquires a lock on buffer, insert the item into empty slot and releases lock on buffer.

P2: Checks whether the buffer has at least one filled slot (if not it will wait to get at least one slot to be filled and will be blocked till then), if found filled slot, it acquires a lock on buffer, removes the item from filled slot and releases lock on buffer.

II. Define the term **spin lock**, what are the **reasons** for a process to be spin locked, is there any way to avoid spin locking, **if yes please mention**? Explain the term **priority inversion** in brief. Consider a non-negative counting semaphore S. The operation P(S) decrements S and V(S) increments S. During an execution the value of S is 7 and 17 P(S) operations and 10 V(S) operations are issued in some order. What can be the resultant value of S after all operations are performed?

III. What is Semaphore explain with example and its type. For the following code executions determine **how many maximum processes** can be in critical section at a time. Assuming initial value of Semaphore variable S is 1.

P1	P2	P3	P4
{	{	{	{
Wait(S)	**Signal(S)**	Wait(S)	Wait(S)
//CS	**//CS**	//CS	//CS
Signal(S)	**Wait(S)**	Signal(S)	Signal(S)
}	}	}	}

Where wait(S) and signal(S) has their usual definitions

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Second Mid-Term Examination, Odd Semester 2019-20

B.Tech(CSE), II Year, III Semester

BCSC0005: Computer Organization

Time : 1 Hour **Maximum Marks: 15**

Section – A

Note: All questions are compulsory **3 x 2 = 6 Marks**

Q1.Explain in brief the control unit design of Microinstructions with next address field.

Q2.Convert the following numerical arithmetic expression into reverse polish notation and show the stack operations for evaluating the numerical result

$$3 + 4 * [(4 + 7 * 7) + 5] * 8 + 2$$

Q3.The 8-bit registers AR, BR, CR and DR initially have the following values:

AR = 11110010 ;BR = 11111111 ; CR = 10111001 ; DR = 11101010

Determine the 8-bit updated values in the register after the execution of the following sequence of micro-operations:

AR ← AR+ BR
CR ← CR ∨ DR, BR← BR + 1
AR ← AR ∧ CR

Section – B

Note: All questions are compulsory **3 x 3 = 9 Marks**

Q1. Write a program to evaluate the following arithmetic statement by using a general register computer with TWO address and ONE Address instruction.

$$X = A - B + C * (D * E - F) / G + H * K$$

Q2. Give one key difference between Hardwired and Microprogrammed control unit. Write a complete sequence of control steps required for the execution of the instruction **MUL R1,R2** by using single bus organization CPU.

Q3. In certain scientific computation it is necessary to perform the arithmetic operation $A_i * B_i + C_i * D_i$ with a stream of numbers. Specify a pipeline configuration to carry out this task. What is speedup ratio and maximum speedup with respect to any pipeline system?

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Second Mid Term Examination, Odd Semester 2019 – 20
B. Tech. II Year III Semester
Engineering Mathematics III (BMAS 0103)

Time: 1 Hr. **Total Marks: 15**

SECTION A **(2 x 3 = 6 marks)**

Note: Attempt **ALL** questions. The symbols have their usual meanings.

Q.1. Find Laplace transforms of

(a) $F(t) = \int_0^t \int_0^t \int_0^t \cos au \, du du du$ (b) $F(t) = t^{7/2} e^{3t}$

Q.2. Find the Laplace transform of $\frac{1-\cos at}{t}$

Q.3. Use convolution theorem to find the inverse Laplace transforms of

$$\frac{1}{p^2(p^2+1)}$$

SECTION B **(3 x 3 = 9 marks)**

Note: Attempt **ALL** questions. The symbols have their usual meanings.

Q.1. Solve the following differential equation using Laplace transforms:

$$y'' - 2y' - 8y = 0$$

with initial conditions, y (0) = 3, $y'(0) = 6$.

(Dash denotes differentiation with respect to t)

Q.2. Evaluate: $L^{-1}\left[\frac{5p+3}{(p-1)(p^2+2p+5)}\right]$

Q.3. Find L [F (t)] where F (t) is a periodic function given by:

$$F(t) = \begin{cases} 1, & 0 \le t < 2 \\ -1, & 2 \le t < 4 \end{cases}$$

It is given that, F (t + 4) = F (t)

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Second Mid-Term Examination, Odd Semester 2019-20

B.Tech (CSE/CCV/DA/IoT/CSF), IInd Year, IIIrd Semester

BCSC0002 : Object-Oriented Programming

Time : 1 Hour Maximum Marks: 15

Section – A

Three questions of 02 marks each. 3 x 2 = 6 Marks

Q1)

a) Define auto-widening, auto-upcasting, and auto-boxing?

b) Write a program to print the maximum and minimum values can be stored in object reference of all the applicable wrapper classes.

Q2)

a) Explain situation when StringBuffer or StringBuilder would be used for string handling.

b) Write a program which appends the string "programming" to string "java". Print the final content of the appended string.

Q3) Explain the following :-

a) Exception and Error
b) throw and throws
c) Checked and Unchecked Exceptions
d) Suppose s1 and s2 are two strings. What is the result of the following code?

```
System.out.println(s1.equals(s2) == s2.equals(s1));
```

Section – B

Three questions of 03 marks each.(Part a 1 Marks Part b 2 Marks) 3 x 3 = 9 Marks

Q1)

a) **Write a program to define custom exception "MarksOutOfBoundException", which when generated will print "Marks is >100".**

b) **why do we need Generics? Give an example of how Generics make a program more flexible?**

Q2)

a) **What is multithreading? Explain two advantages of multithreaded programs.**

b) **Write a program in java to explain how different priorities can be assigned to different threads.**

Q3) Create a list of Employee where employee class contains private fields (int empid, String name, int salary). Sort the list on the basis of name using the method sort() of Collection interface.

Printed Pages: 1 **University Roll No.....**

SecondMid-Term Examination, Odd Semester 2019-20

B.Tech. (ECE), Third, Fifth

BCSO 0002: Introduction to Object Oriented Programming

Time: 1 Hour **Maximum Marks: 15**

Section – A

Attempt all the questions **3 x 2 = 6 Marks**

1. Explain the role of `super` keyword in Java with code snippet.
2. Is multiple inheritance supported in Java? Explain with an example.
3. What is the difference between Abstract class and Interface in Java?

Section – B

Attempt all the questions **3 x 3 = 9 Marks**

1. Write a java program to create an abstract class named Shape that contains an abstractmethod `printArea()`. Provide three classes named `Rectangle`, `Triangle` and `Circle` such that each one of the classes extends the class Shape and overrides the method `printArea()` that prints the area of the given shape.
   Explain how dynamic dispatching can be implemented in the given use case.

2. What is Thread in Java? How can we create threads in Java? Explain the life cycle of Thread

3. Explain the following in context to exception handling in Java:
   i) Try-catch
   ii) Finally
   iii) Throw
   iv) Throws

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Second Mid-Term Examination, Odd Semester 2019-20

B. Tech (CSE), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour

Maximum Marks: 15

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X6=6)

According to moral desert argument, what one deserves is based on his or her merit, skills and efforts. In ancient times, Aristotle made this point clear that justice is giving people what they deserve. But how do we know what people deserve? What goods and opportunities should go to which persons? Aristotle's answer is that we have to consider the telos—the point, the end, or the purpose—of the good in question. Say we have some nice flutes. Who should get them? According to Aristotle, it's not the rich person, since playing flutes has nothing to do with money, nor is it the person who will be made most happy, since making good music is different than being happy. The purpose of a flute is to be played and to be played well. Therefore, Aristotle thinks, the flutes should go to the best flute players. Aristotle's method is to think about justice by thinking about the purpose of a good, an institution, or even a person. If the purpose of a tennis court is to play tennis, then the best tennis players should get priority. If the purpose of universities is to create knowledge and to foster excellence, then the best students should be admitted and the best scientists and professors should be allowed to work at them. If the purpose of being human is to live a good life, then society should promote the good life by ensuring that citizens have the resources necessary for living a good life and by encouraging them in the pursuits that make for a good life. Is this the right way to think about justice?

Questions:

1. We should respect the virtues. .Which of the following points are **inconsistent** with this Aristotelian view?

1. The best flutes must get best flute players.
2. We should distribute according to merits.
3. The best flute players morally deserve to get best flute.
4. The best flute must go to one who can best utilize it for social good.
5. The best flutes must go to who can afford it.
6. The best flutes must go to one who is interested.

a) 4,5,6 b) 1,2&4 c)1,2&3 d)1,2,3,4,5 & 6

2. What is the teleological view of justice?
   1. It is to create a society in which everybody is equal.
   2. A person or an institution deserves according to their purpose.
   3. It is to ensure that rights are not violated.
   4. It is to fulfill everyone's need.
   5. It is to create a better society.
   6. It is according to moral desert.

   a) 1,2 & 3 b)4,5 & 6 c)2,5&6 d)only 2

3. Which of the following statements can be said to be the purpose of a university according to Aristotle?
   1. It is to provide a good placement.
   2. It is to give admission to the needy people.
   3. It is to admit best students.
   4. It is to allow best scientists and professors to work.
   5. It is to develop virtues.
   6. It is to shape good character.

   a)1,2 ,3 & 6 b) 2,3 ,4 & 5 c) 1, 5 & 6d) 3,4,5&6

4. Which of the following are questions about the Aristotelian view of justice?
   1. How do we know what people need?
   2. What goods and opportunities should go to which persons?
   3. Who deserves what?
   4. What is the best utility of a resource in distribution?
   5. What is the purpose of justice?
   6. All above.

   a) 1, 2, 3 & 4 b) 2, 3 & 5 c) only 6 d) 1, 2, 3 & 5

5. Which of the following statements are **wrong** about justice according to Aristotle?
   1. Justice is based on moral desert.
   2. All social institutions should fulfill their purpose.
   3. Justice is equality in society.
   4. It is based on respect for a human being's needs.
   5. It is based on pleasure and pain.
   6. It is based on to achieve maximum good.

   a) 3, 4, 5&6 b) 1, 2 &3 c) only 1& 6 d) only 1, 3 & 6

6. The purpose of justice is to create a better society according to Aristotle. What we can conclude from this statement?
   1. Society should mind the maximum good for all.
   2. Justice is about social justice taking care of all irrespective of the merits.
   3. Better society can be formed only by best people doing for profit.
   4. Only dutiful people can create better society.
   5. Only powerful people can create a better society.
   6. None of the above.

   a)1,2,3,4& 5 b)2,3,4&5 c) only 4,5& 6 d) only 6

Passage: 2 (10X.5= 5)

Sir Isaac Newton talks about the laws of nature: Universal, necessary and certain laws by which the physical world is determined. Impressed by him Kant tried to lay down the foundation of morality with the same universality necessity and uniformity. Kant argues that moral actions, rules and principles have their own moral worth and their moral worth is not dependent on either the desires or the results of actions. He says we act morally only when we act rationally and with a sense of duty. Hence, performing moral actions for Kant is to act out of respect for duty. Let's take an example. Suppose if you are asked why do students work hard in their academics? Most of the students would answer: it is to get good grades or marks in the exam. However, Kant would object to this motivation for working hard in the student life because he would say that your action is not governed by the right kind of motive. Your action does not follow the right reason. Moreover, when you act with this motive, you are not free. The result, which is your motive, is not within your control. You have control only on your actions and on that you can reason, plan and perform. A similar line of reasoning we find in the Shrimad Bhagwad Geeta where Lord Krishna tries to convince Arjuna to fight the war of Mahabharata without being influenced by feelings or results because being a warrior, it was his duty to do so. So, the right reason for working hard in academics would be that being a student, it is one's duty to learn and gain knowledge. Good grades, success and job are mere outcomes. For Kant morality is based on rationality and through rationality we can have freedom to perform our actions. In this way Kant puts emphasis on the sense of duty and rejects the sense of our desires or inclinations as a source of right reason for moral actions. For Kant moral actions are those which follow moral laws. Kant further differentiates between a Moral law and a maxim. Moral law is universally applicable to all human beings. However, a maxim is moral rule or principle that an individual gives to one-self tin the course of reasoning about the morality of an action. Kant further says that there should be Universalizability in the maxims we give to ourselves. What it means is that we should act only on a maxim through which one can at the same time accept that the maxim could be adopted by everyone thereby making it a moral law. This sort of universal agreement grounded in rationality becomes the mark of morality for the Kant. If something is morally right then it has to have the universal acceptance of humanity. Thereafter, Kant talks about the supreme principle of morality what he calls "Categorical Imperative". According to him the categorical imperative is unconditional and does not depend on the consequent factor, while a hypothetical imperative conditionally depends upon the consequences of an action or the consequent factor. For example, work hard if you want to get the 1st rank in the exam. It is a hypothetical imperative because in this statement the hard work is a means to achieve the 1st rank and it is the purpose of the hard work. Working hard in this sense not guided by the sense of duty but is guided by the sense of results. According to Kant this imperative is conditional and cannot give the moral command for the moral actions. First categorical imperative Kant analyzes is: Universalize your Maxim. By this Categorical Imperative Kant means that your individual rule for your actions has that reasonable motive which can be universalized.

Questions:

7. According to Kant, what is nature of the moral law?

1. Universal
2. Uniform
3. Categorical
4. Based on duty.
5. All above

a)2&4 b)1&2 c) only 5 d)2&3

8. What, among the following, is **right** about the meaning of the word "Maxim"?

1. A maxim is a reasonable principle.
2. A moral rule or principle by which an individual governs his moral actions.
3. A moral rule that is not for an individual.
4. Moral rules that cannot be universalized.

a)1&2 b)2&3 c)only2 d)2&4

9. What is similarity between the Shrimad Bhagwat Geeta and the Kantian theory?

1. Both accept result as the ground for right action.
2. Both argue that actions should be detached from our desires.
3. Both focus on dutifulness.
4. Both views advocate doing actions without considering the consequences.

a)1&4 b)2,3&4 c) 1,2&4 d)1,2&3

10. What is right action according to Kant?

1. An action that is performed with the motive of duty.
2. An action based on self-legislated moral law.
3. An action which is commanded by God
4. An action which is prescribed by the society

a)1&3 b)1,2&4 c)1&2 d)3&4

11. What is a "Hypothetical Imperative"?

1. A conditional moral command.
2. An imperative which is not based on desires.
3. An imperative which is grounded in results.
4. An imperative which has universalizability.
5. All above

a)1 & 3 b)1,2&3 c)2,3&4 d)1,2&4

12. If a teacher likes only those students who score well in his subject. How would you judge him from the Kantian point of view?

1. He is utilitarian.
2. He likes the results only.
3. He should treat everyone equally.
4. He is wrong in his approach.
5. All above.

a)1,3&4 b)1,2&4 c)2&3 d) only 5

13. What is incorrect about the moral law?

1. A moral law is self-imposed.
2. A moral law is based on categorical reason.
3. A moral law has universalizability.
4. A moral law is universal.
5. A maxim is an individual expression of moral law.
6. None of the above.

a)1,3&4 b) only 6 c)2&3& 4 d)1,2&3

14. What is universalizability?

1. Universal acceptance grounded in rationality.
2. Keeping in view the purpose of nature & society.
3. It makes a maxim universal and morally right.
4. Universal disagreement on a reasonable principle.
5. It is minding ones interest.

a)1,3&4 & 5 b)1, 2 ,3 & 4 c) 1& 3 d)1,2&3

15. "It is duty of a student to work hard". What we can conclude about this categorical imperative?

1. This motive is not based on external factors like desires and results.
2. This motive is categorical imperative.
3. This motive is minding the very action.
4. This motive follows the sense of duty.
5. All above

a)1,2,3,4&5 b)2&4 c)1,2&3 d)1&4

16. Which of the following statements are correct about the physical world?

1. It is governed by the laws of nature.
2. It is governed by universal laws.
3. It is governed by freedom.
4. It is governed by determinism.
5. None of the above.

a)1,2,3,4&5 b)2,3 ,4 ,5 c) Only 5 d) 1,2 & 4

Answer the following questions based on the theory of Utilitarianism:

Questions: (4X.5=2)

17. A pleasure is pleasure and a pain is pain what Bentham means by this statement.

1. All pleasures are qualitatively same.
2. All pleasures are quantitatively same.
3. What matters is pleasure only.
4. All pleasures are qualitatively different.
5. None of the above.

a. 1, 2, 3 b. only 5 c. 1 & 3 d. only 1 & 2

18. What is being human being according to Mill?
1. Applying thinking & reasoning.
2. Following traditions.
3. Following pleasures only.
4. Making a choice.
5. All above.

a. 1& 4 b. 1, 2, 3 & 4 c. only 5 d. only 1

19. What is psychological hedonism?
1. We human desire pleasures.
2. We human avoid pains.
3. We human avoid pleasures.
4. We human desire pains.
5. We desire both.
6. We avoid both.

a. 1, 2, 3 4, b. 2, 3, 4 c. 4, 5, 6 d. 1& 2

20. Those that don't make a choice are like chimps. What Mill emphasizes by this statement.
1. They act like animals.
2. They simply follow.
3. They are not free.
4. They don't rationalize.
5. They don't apply the human mind.
6. All above.

a. Only 6 b. Only 1, 2,3 c. only 3,4,5 d. only 2,4 & 5

Following questions are based on the case study ship or not ship (4X.5=2)

Rachel works as a Quality Assurance Engineer at a large electronics company. She is responsible for the final testing of her company's servers and is part of a team which decides when new products will be shipped to distributors for sale. Rachel's company has a contract with another company which makes the chips which are incorporated into the servers Rachel's company makes. The business model for this product is to release a new generation server approximately every six months,

meaning Rachel has a limited timeframe to conduct her Quality Control tests. Because there is such a short amount of time between the release of each next new product, the Quality and Assurance department cannot perform every possible test on the servers to ensure they are defect free. Rachel will not ship a product if there is any possibility that the server could malfunction and cause physical harm to the customer. However, she will ship a product that has a higher likelihood of failure resulting in data loss for the customer, because she knows that if she doesn't, her company's competitor will.

21. What is ethically incorrect to do in this case?

1. I would not mind ethical handling of the product.
2. In the short time I maintain that there should not be physical harm.
3. I would ship the product with least damage to my client.
4. I would simply ship the product as to earn profit.
5. I would simply ship the product.

a. 1,4 & 5 b 1,2,3,4,5 c. 2,3,4,5 d. only 4

22. What would Rachel do in the case?

1. Maintain the quality.
2. Simply mind the competition.
3. She should not think so much.
4. She should work independently on her will.
5. She should care the physical harm to the client.

a. 1,2,3,4 b. Only 3,4 c. only 1 & 5 d. 1,2,4& 6

23. Which of the following statements are **ethically correct** about the case?

1. We must deliver the product without checks.
2. In limited frame of time we should give priority to profits only.
3. We should value companies' interest only.
4. We should mind our job only.
5. Only 2

6 None of the above.

a. Only 2,3 ,4 b. Only 6 c. 1,2,3,4 d. 1,2 ,3

24. What would be right from the professional ethics point of view in the case?

1. Rachell shipping the product with all possible checks.
2. Rachell simply minding the business.
3. Rachell simply shipping before her competitor.
4. Rachell not shipping the product.
5. Rachell not caring the physical harms.
6. Rachell simply caring the profit.

a. 2,3,4,5, 6 b. 1, 2, 3, 4 c. only 1 d. only 6

[SET B]

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University Roll No...

Second Mid-Term Examination, Odd Semester 2019-20

B. Tech (CSE), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour

Maximum Marks: 15

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (10X.5= 5)

Sir Isaac Newton talks about the laws of nature: Universal, necessary and certain laws by which the physical world is determined. Impressed by him Kant tried to lay down the foundation of morality with the same universality necessity and uniformity. Kant argues that moral actions, rules and principles have their own moral worth and their moral worth is not dependent on either the desires or the results of actions. He says we act morally only when we act rationally and with a sense of duty. Hence, performing moral actions for Kant is to act out of respect for duty. Let's take an example. Suppose if you are asked why do students work hard in their academics? Most of the students would answer: it is to get good grades or marks in the exam. However, Kant would object to this motivation for working hard in the student life because he would say that your action is not governed by the right kind of motive. Your action does not follow the right reason. Moreover, when you act with this motive, you are not free. The result, which is your motive, is not within your control. You have control only on your actions and on that you can reason, plan and perform. A similar line of reasoning we find in the Shrimad Bhagwad Geeta where Lord Krishna tries to convince Arjuna to fight the war of Mahabharata without being influenced by feelings or results because being a warrior, it was his duty to do so. So, the right reason for working hard in academics would be that being a student, it is one's duty to learn and gain knowledge. Good grades, success and job are mere outcomes. For Kant morality is based on rationality and through rationality we can have freedom to perform our actions. In this way Kant puts emphasis on the sense of duty and rejects the sense of our desires or inclinations as a source of right reason for moral actions. For Kant moral actions are those which follow moral laws. Kant further differentiates between a Moral law and a maxim. Moral law is universally applicable to all human beings. However, a maxim is moral rule or principle that an individual gives to one-self tin the course of reasoning about the morality of an action. Kant further says that there should be Universalizability in the maxims we give to ourselves. What it means is that we should act only on a maxim through which one can at the same time accept that the maxim could be adopted by everyone thereby making it a moral law. This sort of universal agreement grounded in rationality becomes the mark of morality for the Kant. If something is morally right then it has to have the universal acceptance of humanity.

Thereafter, Kant talks about the supreme principle of morality what he calls "Categorical Imperative". According to him the categorical imperative is unconditional and does not depend on the consequent factor, while a hypothetical imperative conditionally depends upon the consequences of an action or the consequent factor. For example, work hard if you want to get the 1st rank in the exam. It is a hypothetical imperative because in this statement the hard work is a means to achieve the 1st rank and it is the purpose of the hard work. Working hard in this sense not guided by the sense of duty but is guided by the sense of results. According to Kant this imperative is conditional and cannot give the moral command for the moral actions. First categorical imperative Kant analyzes is: Universalize your Maxim. By this Categorical Imperative Kant means that your individual rule for your actions has that reasonable motive which can be universalized.

Questions:

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1. Universal
2. Uniform
3. Categorical
4. Based on duty.
5. All above

a)2&4 b)1&2 c) only 5 d)2&3

2. What, among the following, is right about the meaning of the word "Maxim"?

1. A maxim is a reasonable principle.
2. A moral rule or principle by which an individual governs his moral actions.
3. A moral rule that is not for an individual.
4. Moral rules that cannot be universalized.

a)1&2 b)2&3 c)only2 d)2&4

3. What is similarity between the Shrimad Bhagwat Geeta and the Kantian theory?

1. Both accept result as the ground for right action.
2. Both argue that actions should be detached from our desires.
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4. Both views advocate doing actions without considering the consequences.

a)1&4 b)2,3&4 c) 1,2&4 d)1,2&3

4. What is right action according to Kant?

1. An action that is performed with the motive of duty.
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4. He is wrong in his approach.
5. All above.

a)1,3&4 b)1,2&4 c)2&3 d) only 5

7. What is incorrect about the moral law?

1. A moral law is self-imposed.
2. A moral law is based on categorical reason.
3. A moral law has universalizability.
4. A moral law is universal.
5. A maxim is an individual expression of moral law.
6. None of the above.

a)1,3&4 b) only 6 c)2&3& 4 d)1,2&3

8. What is universalizability?

1. Universal acceptance grounded in rationality.
2. Keeping in view the purpose of nature & society.
3. It makes a maxim universal and morally right.
4. Universal disagreement on a reasonable principle.
5. It is minding ones interest.

a)1,3&4 & 5 b)1, 2 ,3 & 4 c) 1& 3 d)1,2&3

9. "It is duty of a student to work hard". What we can conclude about this categorical imperative?

1. This motive is not based on external factors like desires and results.
2. This motive is categorical imperative.
3. This motive is minding the very action.
4. This motive follows the sense of duty.
5. All above

a)1,2,3,4&5 b)2&4 c)1,2&3 d)1&4

10. Which of the following statements are correct about the physical world?

1. It is governed by the laws of nature.
2. It is governed by universal laws.
3. It is governed by freedom.
4. It is governed by determinism.
5. None of the above.

a)1,2,3,4&5 b)2,3 ,4 ,5 c) Only 5 d) 1,2 & 4

Passage: 2 (1X6=6)

According to moral desert argument, what one deserves is based on his or her merit, skills and efforts. In ancient times, Aristotle made this point clear that justice is giving people what they deserve. But how do we know what people deserve? What goods and opportunities should go to which persons? Aristotle's answer is that we have to consider the telos—the point, the end, or the purpose—of the good in question. Say we have some nice flutes. Who should get them? According to Aristotle, it's not the rich person, since playing flutes has nothing to do with money, nor is it the person who will be made most happy, since making good music is different than being happy. The purpose of a flute is to be played and to be played well. Therefore, Aristotle thinks, the flutes should go to the best flute players. Aristotle's method is to think about justice by thinking about the purpose of a good, an institution, or even a person. If the purpose of a tennis court is to play tennis, then the best tennis players should get priority. If the purpose of universities is to create knowledge and to foster excellence, then the best students should be admitted and the best scientists and professors should be allowed to work at them. If the purpose of being human is to live a good life, then society should promote the good life by ensuring that citizens have the resources necessary for living a good life and by encouraging them in the pursuits that make for a good life. Is this the right way to think about justice?

Questions:

11. We should respect the virtues. Which of the following points are **inconsistent** with this Aristotelian view?

1. The best flutes must get best flute players.
2. We should distribute according to merits.
3. The best flute players morally deserve to get best flute.
4. The best flute must go to one who can best utilize it for social good.
5. The best flutes must go to who can afford it.
6. The best flutes must go to one who is interested.

a) 4,5,6 b) 1,2&4 c)1,2&3 d)1,2,3,4,5 & 6

12. What is the teleological view of justice?

1. It is to create a society in which everybody is equal.
2. A person or an institution deserves according to their purpose.
3. It is to ensure that rights are not violated.
4. It is to fulfill everyone's need.
5. It is to create a better society.
6. It is according to moral desert.

a) 1,2 & 3 b)4,5 & 6 c)2,5&6 d)only 2

13. Which of the following statements can be said to be the purpose of a university according to Aristotle?

1. It is to provide a good placement.
2. It is to give admission to the needy people.
3. It is to admit best students.
4. It is to allow best scientists and professors to work.
5. It is to develop virtues.
6. It is to shape good character.

a)1,2 ,3 & 6 b) 2,3 ,4 & 5 c) 1, 5 & 6d) 3,4,5&6

14. Which of the following are questions about the Aristotelian view of justice?

1. How do we know what people need?
2. What goods and opportunities should go to which persons?
3. Who deserves what?
4. What is the best utility of a resource in distribution?
5. What is the purpose of justice?
6. All above.

a) 1, 2, 3 & 4 b) 2, 3 & 5 c) only 6 d) 1, 2, 3 & 5

15. Which of the following statements are **wrong** about justice according to Aristotle?

1. Justice is based on moral desert.
2. All social institutions should fulfill their purpose.
3. Justice is equality in society.
4. It is based on respect for a human being's needs.
5. It is based on pleasure and pain.
6. It is based on to achieve maximum good.

a) 3, 4, 5&6 b) 1, 2 &3 c) only 1& 6 d) only 1, 3 & 6

16. The purpose of justice is to create a better society according to Aristotle. What we can conclude from this statement?

1. Society should mind the maximum good for all.
2. Justice is about social justice taking care of all irrespective of the merits.
3. Better society can be formed only by best people doing for profit.
4. Only dutiful people can create better society.
5. Only powerful people can create a better society.
6. None of the above.

a)1,2,3,4& 5 b)2,3,4&5 c) only 4,5& 6 d) only 6

Answer the following questions based on the theory of Utilitarianism:

Questions: (4X.5=2)

17. A pleasure is pleasure and a pain is pain what Bentham means by this statement.

1. All pleasures are qualitatively same.
2. All pleasures are quantitatively same.
3. What matters is pleasure only.
4. All pleasures are qualitatively different.
5. None of the above.

a. 1, 2, 3 b. only 5 c. 1 & 3 d. only 1 & 2

18. What is being human being according to Mill?

1. Applying thinking & reasoning.
2. Following traditions.
3. Following pleasures only.
4. Making a choice.
5. All above.

a. 1& 4 b. 1, 2, 3 & 4 c. only 5 d. only 1

19. What is psychological hedonism?

1. We human desire pleasures.
2. We human avoid pains.
3. We human avoid pleasures.
4. We human desire pains.
5. We desire both.
6. We avoid both.

a. 1, 2, 3 4, b. 2, 3, 4 c. 4, 5, 6 d. 1& 2

20. Those that don't make a choice are like chimps. What Mill emphasizes by this statement.

1. They act like animals.
2. They simply follow.
3. They are not free.
4. They don't rationalize.
5. They don't apply the human mind.
6. All above.

a. Only 6 b. Only 1, 2,3 c. only 3,4,5 d. only 2,4 & 5

Following questions are based on the case study ship or not ship (4X.5=2)

Rachel works as a Quality Assurance Engineer at a large electronics company. She is responsible for the final testing of her company's servers and is part of a team which decides when new products will be shipped to distributors for sale. Rachel's company has a contract with another company which makes the chips which are incorporated into the servers Rachel's company makes. The business model for this product is to release a new generation server approximately every six months, meaning Rachel has a limited timeframe to conduct her Quality Control tests. Because there is such a short amount of time between the release of each next new product, the Quality and Assurance department cannot perform every possible test on the servers to ensure they are defect free. Rachel will not ship a product if there is any possibility that the server could malfunction and cause physical harm to the customer. However, she will ship a product that has a higher likelihood of failure resulting in data loss for the customer, because she knows that if she doesn't, her company's competitor will.

21. What is ethically incorrect to do in this case?

1. I would not mind ethical handling of the product.
2. In the short time I maintain that there should not be physical harm.
3. I would ship the product with least damage to my client.
4. I would simply ship the product as to earn profit.
5. I would simply ship the product.

a. 1,4 & 5 **b 1,2,3,4,5 c. 2,3,4,5 d. only 4**

22. What would Rachel do in the case?

1. Maintain the quality.
2. Simply mind the competition.
3. She should not think so much.
4. She should work independently on her will.
5. She should care the physical harm to the client.

a. 1,2,3,4 b. Only 3,4 c. only 1 & 5 d. 1,2,4& 6

23. Which of the following statements are **ethically correct** about the case?

1. We must deliver the product without checks.
2. In limited frame of time we should give priority to profits only.
3. We should value companies' interest only.
4. We should mind our job only.
5. Only 2
6 None of the above.

a. Only 2,3 ,4 b. Only 6 c. 1,2,3,4 d. 1,2 ,3

24. What would be right from the professional ethics point of view in the case?

1. Rachell shipping the product with all possible checks.
2. Rachell simply minding the business.
3. Rachell simply shipping before her competitor.
4. Rachell not shipping the product.
5. Rachell not caring the physical harms.
6. Rachell simply caring the profit.

a. 2,3,4,5, 6 b. 1, 2, 3, 4 c. only 1 d. only 6

[SET C]

Printed Pages: 8 **University Roll No...**

Second Mid-Term Examination, Odd Semester 2019-20

B. Tech (CSE), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour **Maximum Marks: 15**

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X6=6)

According to moral desert argument, what one deserves is based on his or her merit, skills and efforts. In ancient times, Aristotle made this point clear that justice is giving people what they deserve. But how do we know what people deserve? What goods and opportunities should go to which persons? Aristotle's answer is that we have to consider the telos—the point, the end, or the purpose—of the good in question. Say we have some nice flutes. Who should get them? According to Aristotle, it's not the rich person, since playing flutes has nothing to do with money, nor is it the person who will be made most happy, since making good music is different than being happy. The purpose of a flute is to be played and to be played well. Therefore, Aristotle thinks, the flutes should go to the best flute players. Aristotle's method is to think about justice by thinking about the purpose of a good, an institution, or even a person. If the purpose of a tennis court is to play tennis, then the best tennis players should get priority. If the purpose of universities is to create knowledge and to foster excellence, then the best students should be admitted and the best scientists and professors should be allowed to work at them. If the purpose of being human is to live a good life, then society should promote the good life by ensuring that citizens have the resources necessary for living a good life and by encouraging them in the pursuits that make for a good life. Is this the right way to think about justice?

Questions:

1. We should respect the virtues. Which of the following points are **inconsistent** with this Aristotelian view?
   1. The best flutes must get best flute players.
   2. We should distribute according to merits.
   3. The best flute players morally deserve to get best flute.
   4. The best flute must go to one who can best utilize it for social good.
   5. The best flutes must go to who can afford it.
   6. The best flutes must go to one who is interested.

   a) 4,5,6 b) 1,2&4 c)1,2&3 d)1,2,3,4,5 & 6

2. What is the teleological view of justice?
   1. It is to create a society in which everybody is equal.
   2. A person or an institution deserves according to their purpose.
   3. It is to ensure that rights are not violated.
   4. It is to fulfill everyone's need.
   5. It is to create a better society.
   6. It is according to moral desert.

   a) 1,2 & 3 b)4,5 & 6 c)2,5&6 d)only 2

3. Which of the following statements can be said to be the purpose of a university according to Aristotle?
   1. It is to provide a good placement.
   2. It is to give admission to the needy people.
   3. It is to admit best students.
   4. It is to allow best scientists and professors to work.
   5. It is to develop virtues.
   6. It is to shape good character.

   a)1,2 ,3 & 6 b) 2,3 ,4 & 5 c) 1, 5 & 6d) 3,4,5&6

4. Which of the following are questions about the Aristotelian view of justice?
   1. How do we know what people need?
   2. What goods and opportunities should go to which persons?
   3. Who deserves what?
   4. What is the best utility of a resource in distribution?
   5. What is the purpose of justice?
   6. All above.

   a) 1, 2, 3 & 4 b) 2, 3 & 5 c) only 6 d) 1, 2, 3 & 5

5. Which of the following statements are **wrong** about justice according to Aristotle?
   1. Justice is based on moral desert.
   2. All social institutions should fulfill their purpose.
   3. Justice is equality in society.
   4. It is based on respect for a human being's needs.
   5. It is based on pleasure and pain.
   6. It is based on to achieve maximum good.

   a) 3, 4, 5&6 b) 1, 2 &3 c) only 1& 6 d) only 1, 3 & 6

6. The purpose of justice is to create a better society according to Aristotle. What we can conclude from this statement?
   1. Society should mind the maximum good for all.
   2. Justice is about social justice taking care of all irrespective of the merits.
   3. Better society can be formed only by best people doing for profit.
   4. Only dutiful people can create better society.
   5. Only powerful people can create a better society.
   6. None of the above.

   a)1,2,3,4& 5 b)2,3,4&5 c) only 4,5& 6 d) only 6

Passage: 2 (10X.5= 5)

Sir Isaac Newton talks about the laws of nature: Universal, necessary and certain laws by which the physical world is determined. Impressed by him Kant tried to lay down the foundation of morality with the same universality necessity and uniformity. Kant argues that moral actions, rules and principles have their own moral worth and their moral worth is not dependent on either the desires or the results of actions. He says we act morally only when we act rationally and with a sense of duty. Hence, performing moral actions for Kant is to act out of respect for duty. Let's take an example. Suppose if you are asked why do students work hard in their academics? Most of the students would answer: it is to get good grades or marks in the exam. However, Kant would object to this motivation for working hard in the student life because he would say that your action is not governed by the right kind of motive. Your action does not follow the right reason. Moreover, when you act with this motive, you are not free. The result, which is your motive, is not within your control. You have control only on your actions and on that you can reason, plan and perform. A similar line of reasoning we find in the Shrimad Bhagwad Geeta where Lord Krishna tries to convince Arjuna to fight the war of Mahabharata without being influenced by feelings or results because being a warrior, it was his duty to do so. So, the right reason for working hard in academics would be that being a student, it is one's duty to learn and gain knowledge. Good grades, success and job are mere outcomes. For Kant morality is based on rationality and through rationality we can have freedom to perform our actions. In this way Kant puts emphasis on the sense of duty and rejects the sense of our desires or inclinations as a source of right reason for moral actions. For Kant moral actions are those which follow moral laws. Kant further differentiates between a Moral law and a maxim. Moral law is universally applicable to all human beings. However, a maxim is moral rule or principle that an individual gives to one-self tin the course of reasoning about the morality of an action. Kant further says that there should be Universalizability in the maxims we give to ourselves. What it means is that we should act only on a maxim through which one can at the same time accept that the maxim could be adopted by everyone thereby making it a moral law. This sort of universal agreement grounded in rationality becomes the mark of morality for the Kant. If something is morally right then it has to have the universal acceptance of humanity. Thereafter, Kant talks about the supreme principle of morality what he calls "Categorical Imperative". According to him the categorical imperative is unconditional and does not depend on the consequent factor, while a hypothetical imperative conditionally depends upon the consequences of an action or the consequent factor. For example, work hard if you want to get the 1st rank in the exam. It is a hypothetical imperative because in this statement the hard work is a means to achieve the 1st rank and it is the purpose of the hard work. Working hard in this sense not guided by the sense of duty but is guided by the sense of results. According to Kant this imperative is conditional and cannot give the moral command for the moral actions. First categorical imperative Kant analyzes is: Universalize your Maxim. By this Categorical Imperative Kant means that your individual rule for your actions has that reasonable motive which can be universalized.

Questions:

7. According to Kant, what is nature of the moral law?

1. Universal
2. Uniform
3. Categorical
4. Based on duty.
5. All above

a)2&4 b)1&2 c) only 5 d)2&3

8. What, among the following, is right about the meaning of the word "Maxim"?

1. A maxim is a reasonable principle.
2. A moral rule or principle by which an individual governs his moral actions.
3. A moral rule that is not for an individual.
4. Moral rules that cannot be universalized.

a)1&2 b)2&3 c)only2 d)2&4

9. What is similarity between the Shrimad Bhagwat Geeta and the Kantian theory?

1. Both accept result as the ground for right action.
2. Both argue that actions should be detached from our desires.
3. Both focus on dutifulness.
4. Both views advocate doing actions without considering the consequences.

a)1&4 b)2,3&4 c) 1,2&4 d)1,2&3

10. What is right action according to Kant?

1. An action that is performed with the motive of duty.
2. An action based on self-legislated moral law.
3. An action which is commanded by God
4. An action which is prescribed by the society

a)1&3 b)1,2&4 c)1&2 d)3&4

11. What is a "Hypothetical Imperative"?

1. A conditional moral command.
2. An imperative which is not based on desires.
3. An imperative which is grounded in results.
4. An imperative which has universalizability.
5. All above

a)1 & 3 b)1,2&3 c)2,3&4 d)1,2&4

12. If a teacher likes only those students who score well in his subject. How would you judge him from the Kantian point of view?

1. He is utilitarian.
2. He likes the results only.
3. He should treat everyone equally.
4. He is wrong in his approach.
5. All above.

a)1,3&4 b)1,2&4 c)2&3 d) only 5

13. What is incorrect about the moral law?

1. A moral law is self-imposed.
2. A moral law is based on categorical reason.
3. A moral law has universalizability.
4. A moral law is universal.
5. A maxim is an individual expression of moral law.
6. None of the above.

a)1,3&4 b) only 6 c)2&3& 4 d)1,2&3

14. What is universalizability?

1. Universal acceptance grounded in rationality.
2. Keeping in view the purpose of nature & society.
3. It makes a maxim universal and morally right.
4. Universal disagreement on a reasonable principle.
5. It is minding ones interest.

a)1,3&4 & 5 b)1, 2 ,3 & 4 c) 1& 3 d)1,2&3

15. "It is duty of a student to work hard". What we can conclude about this categorical imperative?

1. This motive is not based on external factors like desires and results.
2. This motive is categorical imperative.
3. This motive is minding the very action.
4. This motive follows the sense of duty.
5. All above

a)1,2,3,4&5 b)2&4 c)1,2&3 d)1&4

16. Which of the following statements are correct about the physical world?

1. It is governed by the laws of nature.
2. It is governed by universal laws.
3. It is governed by freedom.
4. It is governed by determinism.
5. None of the above.

a)1,2,3,4&5 b)2,3 ,4 ,5 c) Only 5 d) 1,2 & 4

Following questions are based on the case study ship or not ship (4X.5=2)

Rachel works as a Quality Assurance Engineer at a large electronics company. She is responsible for the final testing of her company's servers and is part of a team which decides when new products will be shipped to distributors for sale. Rachel's company has a contract with another company which makes the chips which are incorporated into the servers Rachel's company makes. The business model for this product is to release a new generation server approximately every six months, meaning Rachel has a limited timeframe to conduct her Quality Control

tests. Because there is such a short amount of time between the release of each next new product, the Quality and Assurance department cannot perform every possible test on the servers to ensure they are defect free. Rachel will not ship a product if there is any possibility that the server could malfunction and cause physical harm to the customer. However, she will ship a product that has a higher likelihood of failure resulting in data loss for the customer, because she knows that if she doesn't, her company's competitor will.

17. What is ethically incorrect to do in this case?

1. I would not mind ethical handling of the product.
2. In the short time I maintain that there should not be physical harm.
3. I would ship the product with least damage to my client.
4. I would simply ship the product as to earn profit.
5. I would simply ship the product.

a. 1,4 & 5 **b 1,2,3,4,5 c. 2,3,4,5 d. only 4**

18. What would Rachel do in the case?

1. Maintain the quality.
2. Simply mind the competition.
3. She should not think so much.
4. She should work independently on her will.
5. She should care the physical harm to the client.

a. 1,2,3,4 b. Only 3,4 c. only 1 & 5 d. 1,2,4& 6

19. Which of the following statements are **ethically correct** about the case?

1. We must deliver the product without checks.
2. In limited frame of time we should give priority to profits only.
3. We should value companies' interest only.
4. We should mind our job only.
5. Only 2

6 None of the above.

a. Only 2,3 ,4 b. Only 6 c. 1,2,3,4 d. 1,2 ,3

20. What would be right from the professional ethics point of view in the case?

1. Rachell shipping the product with all possible checks.
2. Rachell simply minding the business.
3. Rachell simply shipping before her competitor.
4. Rachell not shipping the product.
5. Rachell not caring the physical harms.
6. Rachell simply caring the profit.

a. 2,3,4,5, 6 b. 1, 2, 3, 4 c. only 1 d. only 6

Answer the following questions based on the theory of Utilitarianism:

Questions: (4X.5=2)

21. A pleasure is pleasure and a pain is pain what Bentham means by this statement.

1. All pleasures are qualitatively same.
2. All pleasures are quantitatively same.
3. What matters is pleasure only.
4. All pleasures are qualitatively different.
5. None of the above.

a. 1, 2, 3 b. only 5 c. 1 & 3 d. only 1 & 2

22. What is being human being according to Mill?

1. Applying thinking & reasoning.
2. Following traditions.
3. Following pleasures only.
4. Making a choice.
5. All above.

a. 1& 4 b. 1, 2, 3 & 4 c. only 5 d. only 1

23. What is psychological hedonism?

1. We human desire pleasures.
2. We human avoid pains.
3. We human avoid pleasures.
4. We human desire pains.
5. We desire both.

6. We avoid both.

a. 1, 2, 3 4, b. 2, 3, 4 c. 4, 5, 6 d. 1& 2

24. Those that don't make a choice are like chimps. What Mill emphasizes by this statement.

1. They act like animals.
2. They simply follow.
3. They are not free.
4. They don't rationalize.
5. They don't apply the human mind.
6. All above.

a. Only 6 b. Only 1, 2,3 c. only 3,4,5 d. only 2,4 & 5

[SET D]

Printed Pages: 8 University Roll No...

Second Mid-Term Examination, Odd Semester 2019-20

B. Tech (CSE), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour **Maximum Marks: 15**

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X6=6)

According to moral desert argument, what one deserves is based on his or her merit, skills and efforts. In ancient times, Aristotle made this point clear that justice is giving people what they deserve. But how do we know what people deserve? What goods and opportunities should go to which persons? Aristotle's answer is that we have to consider the telos—the point, the end, or the purpose—of the good in question. Say we have some nice flutes. Who should get them? According to Aristotle, it's not the rich person, since playing flutes has nothing to do with money, nor is it the person who will be made most happy, since making good music is different than being happy. The purpose of a flute is to be played and to be played well. Therefore, Aristotle thinks, the flutes should go to the best flute players. Aristotle's method is to think about justice by thinking about the purpose of a good, an institution, or even a person. If the purpose of a tennis court is to play tennis, then the best tennis players should get priority. If the purpose of universities is to create knowledge and to foster excellence, then the best students should be admitted and the best scientists and professors should be allowed to work at them. If the purpose of being human is to live a good life, then society should promote the good life by ensuring that citizens have the resources necessary for living a good life and by encouraging them in the pursuits that make for a good life. Is this the right way to think about justice?

Questions:

1. We should respect the virtues. Which of the following points are **inconsistent** with this Aristotelian view?

1. The best flutes must get best flute players.
2. We should distribute according to merits.
3. The best flute players morally deserve to get best flute.
4. The best flute must go to one who can best utilize it for social good.
5. The best flutes must go to who can afford it.
6. The best flutes must go to one who is interested.

a) 4,5,6 b) 1,2&4 c)1,2&3 d)1,2,3,4,5 & 6

2. What is the teleological view of justice?
   1. It is to create a society in which everybody is equal.
   2. A person or an institution deserves according to their purpose.
   3. It is to ensure that rights are not violated.
   4. It is to fulfill everyone's need.
   5. It is to create a better society.
   6. It is according to moral desert.

   a) 1,2 & 3 b)4,5 & 6 c)2,5&6 d)only 2

3. Which of the following statements can be said to be the purpose of a university according to Aristotle?
   1. It is to provide a good placement.
   2. It is to give admission to the needy people.
   3. It is to admit best students.
   4. It is to allow best scientists and professors to work.
   5. It is to develop virtues.
   6. It is to shape good character.

   a)1,2 ,3 & 6 b) 2,3 ,4 & 5 c) 1, 5 & 6d) 3,4,5&6

4. Which of the following are questions about the Aristotelian view of justice?
   1. How do we know what people need?
   2. What goods and opportunities should go to which persons?
   3. Who deserves what?
   4. What is the best utility of a resource in distribution?
   5. What is the purpose of justice?
   6. All above.

   a) 1, 2, 3 & 4 b) 2, 3 & 5 c) only 6 d) 1, 2, 3 & 5

5. Which of the following statements are **wrong** about justice according to Aristotle?
   1. Justice is based on moral desert.
   2. All social institutions should fulfill their purpose.
   3. Justice is equality in society.
   4. It is based on respect for a human being's needs.
   5. It is based on pleasure and pain.
   6. It is based on to achieve maximum good.

   a) 3, 4, 5&6 b) 1, 2 &3 c) only 1& 6 d) only 1, 3 & 6

6. The purpose of justice is to create a better society according to Aristotle. What we can conclude from this statement?
   1. Society should mind the maximum good for all.
   2. Justice is about social justice taking care of all irrespective of the merits.
   3. Better society can be formed only by best people doing for profit.
   4. Only dutiful people can create better society.
   5. Only powerful people can create a better society.
   6. None of the above.

   a)1,2,3,4& 5 b)2,3,4&5 c) only 4,5& 6 d) only 6

Passage: 2 (10X.5= 5)

Sir Isaac Newton talks about the laws of nature: Universal, necessary and certain laws by which the physical world is determined. Impressed by him Kant tried to lay down the foundation of morality with the same universality necessity and uniformity. Kant argues that moral actions, rules and principles have their own moral worth and their moral worth is not dependent on either the desires or the results of actions. He says we act morally only when we act rationally and with a sense of duty. Hence, performing moral actions for Kant is to act out of respect for duty. Let's take an example. Suppose if you are asked why do students work hard in their academics? Most of the students would answer: it is to get good grades or marks in the exam. However, Kant would object to this motivation for working hard in the student life because he would say that your action is not governed by the right kind of motive. Your action does not follow the right reason. Moreover, when you act with this motive, you are not free. The result, which is your motive, is not within your control. You have control only on your actions and on that you can reason, plan and perform. A similar line of reasoning we find in the Shrimad Bhagwad Geeta where Lord Krishna tries to convince Arjuna to fight the war of Mahabharata without being influenced by feelings or results because being a warrior, it was his duty to do so. So, the right reason for working hard in academics would be that being a student, it is one's duty to learn and gain knowledge. Good grades, success and job are mere outcomes. For Kant morality is based on rationality and through rationality we can have freedom to perform our actions. In this way Kant puts emphasis on the sense of duty and rejects the sense of our desires or inclinations as a source of right reason for moral actions. For Kant moral actions are those which follow moral laws. Kant further differentiates between a Moral law and a maxim. Moral law is universally applicable to all human beings. However, a maxim is moral rule or principle that an individual gives to one-self tin the course of reasoning about the morality of an action. Kant further says that there should be Universalizability in the maxims we give to ourselves. What it means is that we should act only on a maxim through which one can at the same time accept that the maxim could be adopted by everyone thereby making it a moral law. This sort of universal agreement grounded in rationality becomes the mark of morality for the Kant. If something is morally right then it has to have the universal acceptance of humanity. Thereafter, Kant talks about the supreme principle of morality what he calls "Categorical Imperative". According to him the categorical imperative is unconditional and does not depend on the consequent factor, while a hypothetical imperative conditionally depends upon the consequences of an action or the consequent factor. For example, work hard if you want to get the 1st rank in the exam. It is a hypothetical imperative because in this statement the hard work is a means to achieve the 1st rank and it is the purpose of the hard work. Working hard in this sense not guided by the sense of duty but is guided by the sense of results. According to Kant this imperative is conditional and cannot give the moral command for the moral actions. First categorical imperative Kant analyzes is: Universalize your Maxim. By this Categorical Imperative Kant means that your individual rule for your actions has that reasonable motive which can be universalized.

Questions:

7. According to Kant, what is nature of the moral law?

1. Universal
2. Uniform
3. Categorical
4. Based on duty.
5. All above

a)2&4 b)1&2 c) only 5 d)2&3

8. What, among the following, is **right** about the meaning of the word "Maxim"?

1. A maxim is a reasonable principle.
2. A moral rule or principle by which an individual governs his moral actions.
3. A moral rule that is not for an individual.
4. Moral rules that cannot be universalized.

a)1&2 b)2&3 c)only2 d)2&4

9. What is similarity between the Shrimad Bhagwat Geeta and the Kantian theory?

1. Both accept result as the ground for right action.
2. Both argue that actions should be detached from our desires.
3. Both focus on dutifulness.
4. Both views advocate doing actions without considering the consequences.

a)1&4 b)2,3&4 c) 1,2&4 d)1,2&3

10. What is right action according to Kant?

1. An action that is performed with the motive of duty.
2. An action based on self-legislated moral law.
3. An action which is commanded by God
4. An action which is prescribed by the society

a)1&3 b)1,2&4 c)1&2 d)3&4

11. What is a "Hypothetical Imperative"?

1. A conditional moral command.
2. An imperative which is not based on desires.
3. An imperative which is grounded in results.
4. An imperative which has universalizability.
5. All above

a)1 & 3 b)1,2&3 c)2,3&4 d)1,2&4

12. If a teacher likes only those students who score well in his subject. How would you judge him from the Kantian point of view?

1. He is utilitarian.
2. He likes the results only.
3. He should treat everyone equally.
4. He is wrong in his approach.
5. All above.

a)1,3&4 b)1,2&4 c)2&3 d) only 5

13. What is incorrect about the moral law?
1. A moral law is self-imposed.
2. A moral law is based on categorical reason.
3. A moral law has universalizability.
4. A moral law is universal.
5. A maxim is an individual expression of moral law.
6. None of the above.

a)1,3&4 b) only 6 c)2&3& 4 d)1,2&3

14. What is universalizability?
1. Universal acceptance grounded in rationality.
2. Keeping in view the purpose of nature & society.
3. It makes a maxim universal and morally right.
4. Universal disagreement on a reasonable principle.
5. It is minding ones interest.

a)1,3&4 & 5 b)1, 2 ,3 & 4 c) 1& 3 d)1,2&3

15. "It is duty of a student to work hard". What we can conclude about this categorical imperative?
1. This motive is not based on external factors like desires and results.
2. This motive is categorical imperative.
3. This motive is minding the very action.
4. This motive follows the sense of duty.
5. All above

a)1,2,3,4&5 b)2&4 c)1,2&3 d)1&4

16. Which of the following statements are correct about the physical world?
1. It is governed by the laws of nature.
2. It is governed by universal laws.
3. It is governed by freedom.
4. It is governed by determinism.
5. None of the above.

a)1,2,3,4&5 b)2,3 ,4 ,5 c) Only 5 d) 1,2 & 4

Answer the following questions based on the theory of Utilitarianism:

Questions: (4X.5=2) .

17. A pleasure is pleasure and a pain is pain what Bentham means by this statement.
1. All pleasures are qualitatively same.
2. All pleasures are quantitatively same.
3. What matters is pleasure only.
4. All pleasures are qualitatively different.
5. None of the above.

a. 1, 2, 3 b. only 5 c. 1 & 3 d. only 1 & 2

18. What is being human being according to Mill?
1. Applying thinking & reasoning.
2. Following traditions.
3. Following pleasures only.
4. Making a choice.
5. All above.

a. 1& 4 b. 1, 2, 3 & 4 c. only 5 d. only 1

19. What is psychological hedonism?
1. We human desire pleasures.
2. We human avoid pains.
3. We human avoid pleasures.
4. We human desire pains.
5. We desire both.
6. We avoid both.

a. 1, 2, 3 4, b. 2, 3, 4 c. 4, 5, 6 d. 1& 2

20. Those that don't make a choice are like chimps. What Mill emphasizes by this statement.
1. They act like animals.
2. They simply follow.
3. They are not free.
4. They don't rationalize.
5. They don't apply the human mind.
6. All above.

a. Only 6 b. Only 1, 2,3 c. only 3,4,5 d. only 2,4 & 5

Following questions are based on the case study ship or not ship (4X.5=2)

Rachel works as a Quality Assurance Engineer at a large electronics company. She is responsible for the final testing of her company's servers and is part of a team which decides when new products will be shipped to distributors for sale. Rachel's company has a contract with another company which makes the chips which are incorporated into the servers Rachel's company makes. The business model for this product is to release a new generation server approximately every six months, meaning Rachel has a limited timeframe to conduct her Quality Control

tests. Because there is such a short amount of time between the release of each next new product, the Quality and Assurance department cannot perform every possible test on the servers to ensure they are defect free. Rachel will not ship a product if there is any possibility that the server could malfunction and cause physical harm to the customer. However, she will ship a product that has a higher likelihood of failure resulting in data loss for the customer, because she knows that if she doesn't, her company's competitor will.

21. What is ethically incorrect to do in this case?

1. I would not mind ethical handling of the product.
2. In the short time I maintain that there should not be physical harm.
3. I would ship the product with least damage to my client.
4. I would simply ship the product as to earn profit.
5. I would simply ship the product.

a. 1,4 & 5 **b 1,2,3,4,5 c. 2,3,4,5 d. only 4**

22. What would Rachel do in the case?

1. Maintain the quality.
2. Simply mind the competition.
3. She should not think so much.
4. She should work independently on her will.
5. She should care the physical harm to the client.

a. 1,2,3,4 b. Only 3,4 c. only 1 & 5 d. 1,2,4& 6

23. Which of the following statements are **ethically correct** about the case?

1. We must deliver the product without checks.
2. In limited frame of time we should give priority to profits only.
3. We should value companies' interest only.
4. We should mind our job only.
5. Only 2
6 None of the above.

a. Only 2,3 ,4 b. Only 6 c. 1,2,3,4 d. 1,2 ,3

24. What would be right from the professional ethics point of view in the case?

1. Rachell shipping the product with all possible checks.
2. Rachell simply minding the business.
3. Rachell simply shipping before her competitor.
4. Rachell not shipping the product.
5. Rachell not caring the physical harms.
6. Rachell simply caring the profit.

a. 2,3,4,5, 6 b. 1, 2, 3, 4 c. only 1 d. only 6

Printed Pages: **University Roll No.....**

Second Mid-Term Examination, Odd Semester 2019-20

Program: B.Tech. (CEA), 2ND Year, 3RD Semester

Subject Code: Database Management System (BCSC0003)

Time: 1 Hour **Maximum Marks: 15**

Section – A

Three questions of 02 marks each (with no internal choice). 3 x 2 = 6 Marks

1. Discuss insertion, deletion and modification anomalies? Illustrate with example.
2. Define prime attribute and candidate key? Suppose we have a database for an investment firm, consisting of the following attributes: (B) Broker, (O) Office of a broker, (I) Investor, (S) Stock, (Q) Quantity of stock owned by an investor, (D) Dividend paid by a stock. With the following functional dependencies: F= {S→D, I→B, I→B, IS→Q, B→O}. Find all the candidate key for the relation schema R (B, O, S, Q, I, D).
3. What is meant by the completeness and soundness of Armstrong's inference rule?

Section – B

Three questions of 03 marks each (with no internal choice). 3 x 3 = 9 Marks

1. Define Boyce-Codd Normal Form. How does it differ from 3NF? why is it considered a stronger form of 3NF.
2. Consider the given relation R (ABCDE) with the set of F = {A→BC, CD→E, E→C, D→AEH, ABH→BD, DH→BC} and compute the following:
   I). Find the Non-Redundant Cover of F.
   II)Also list the candidate keys of R.
3. (a)What are the difference among primary, secondary and clustering indexes? Which of the index are dense, and which are not?
   (b)A relation R(ABCDEFG) and a set of functional dependencies F= {AB→CD, DE→F, C→E, F→C, B→G}. Is it in 2NF? if not then convert into 2NF.

Paper Code-BTDH 0301

SET-C

Soft Skills

Mid Term, Odd Sem Examination (2019-20)

Course- B.tech **Year- II**

Branch- (All)

Time : 60 Min. **M.M-20**

Instructions :

1. **Attempt all questions.**
2. **There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.**
3. **The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.**
4. **The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.**
5. **Only black& blue pen should be used to fill forms.**
6. **Each branch is identified with a specific subject code which should be written in subject column of OMR sheet**
7. **Write your University roll number in the roll no column of OMR sheet.**

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:..... **All The Best**

(Aptitude)-No of question-40, marks-0.50

Q1. A person bought some article at rate of 11 for Rs 10 and bought same number of articles at the rate of 9 for Rs 10. And sold all the stock at the rate of 1 for Rs 1. Find the profit or loss %.
A. 14.28 B. 2.02 C. 6.66 D. 22.22 E. None of these

Q2. A retail offers the following discount scheme for buyers on an articlE. Two successive discounts of 10% is equivalent to single discount of.....
A. 19 B. 16 C. 20 D. 22.22 E. None of these

Q3. Two shopkeepers announce the same price of Rs 700 for a shirt. The first offers successive discounts of 30% and 6% while the second offers successive discounts of 20% and 16%. The shopkeeper that offers better discount is more of
A. 12.4 B. 15 C. 13.6 D. 18 E. None of these

Q4. If selling price of 50 articles is same as cost price 75 articles find profit %.
A. 20 B.30 C.40 D.50 E. None of these

Q5. A businessman sells a commodity at 10% profit. If he had bought it at 10% less and sold it for Rs. 2 less, then he would have gained 50/3%. The cost price of the commodity is
A. Rs. 32 B. Rs. 36 C. Rs. 40 D. Rs. 48 E. None of these

Q6. Lalit marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance also, which reads 1000 gm for 800 gm. What is his net profit percentage?
A. 45 B. 54.75% C. 50% D. 57.5% E. None of these

Q7. Nandlal purchased 20 dozen notebooks at Rs48 per dozen. He sold 8 dozen at 10% profit and the remaining 12 dozen at 20% profit. What is his profit percentage in this transaction?
A. 16 B. 15 C. 10 D. 18 E. None of these

Q8. An articles is sold at 20% loss and another article is sold at 10% gain. find overall profit or loss% if Selling price of both the articles is samE.(approx)
A. 4 B. 8 C. 10 D. 12 E. None of these

Q9. A shopkeeper offers 20% discount and still makes profit of 25%. Calculate the cost of article which has a marked price of Rs 200.
A. Rs. 130 B. Rs. 128 C. Rs. 125 D. Rs. 140
E. None of these

Q10. How many possible pairs of values of (x, y) exist such that the number 42xy60 is divisible by 72?.
A. 1 B. 3 C. 5 D. 7 E. None of these

Q11. If the number 3422213x2 is divisible by 12, find the no. of possible values of x
A. 2 B. 4 C. 5 D. 6 E. None of these

Q12. What will be the day of the week 15th August, 2010?
A.Sunday B.Monday C.Tuesday D.Friday
E. None of these

Q13. Which of the following pair of years have same calendars?
I. 1973 &1979 II. 1974 &1985 III. 1900 & 1928
A.Only I B. Only II C.Only III D.Both I & II
E. Both II & III

Q14.Which of the following can not be last day of 1900?
A.Monday B.Tuesday C.Wednesday D. Friday
E. None of these

Q15. Kishan spends 30% of his salary on food and donates 3% in a Charitable Trust. He spends Rs. 2310 on these two items, then total salary for that month is
A. Rs 6000 B. Rs 8000 C. Rs 9000 D. Rs 7000
E. None of these

Q16. In an examination it is required to get 40% of the aggregate marks to pass. A student gets 522 marks and is declared failed by 4% marks. What are the maximum aggregate marks a student can get?
A. 1700 B. 1730 C. 1450 D. 1765 E. None of these

Q17. A box has 100 blue balls, 50 red balls and 50 black balls, 25% of blue balls and 50% of red balls are taken away. Then, percentage of black balls at present is
A. 25% B. 40% C. 33.33% D. 50% E. None of these

Q18. Two numbers are respectively 10% and 25% more than a third number. What per cent is the first number of the second?
A. 88% B. 65% C. 75% D. 80% E. None of these

Directions for questions 19 to 22: Read the following information carefully and then answer
"In a family there are 6 members A, B, C, D, E and F covering three generations. D has two daughters. A is the eldest member of the family and C is grandmother of E, B is mother of F"

Q19. Who is the youngest member of the family?
A. B B. E C. E & F D. None of these

Q20. What is the related between E and F?
A. Husband -wife B. Mother- daughter C. Father -daughter
D. Sisiter-sister E. None of these

Q21. Who is the wife of D?
A. B B. E C. F D. None of these

Q22. How is A related to E?
A. Grand Father B. Mother C. Sister D. Father
E. None of these

Directions (23-25): Following questions are based on the information provided below:
'**A × B**' means 'A is mother of B'.
'**A – B**' means 'A is brother of B'.
'**A + B**' means 'A is sister of B'.
'**A ÷ B**' means 'A is father of B'.

Q23. If the expression M × N + R ÷ T is true, then which of the following is true?
A. M is father of R B. N is aunt of T C. M is grandfather of T
D. T is sister of N E. None of these

Q24. Which of the following means 'Q' is brother of 'T'?
A. Y ÷ Q – T + R B. T ÷ Q – Y- R C. T × M – R + Q
D. T + M ÷ Y + Q E. None of these

Q25. If the expression M – Q – D ÷ F + C, then which of the following is true?
A. C is daughter of F B. F is niece of C C. C is sister of F
D. D is father of M E. None of these

Q26. Pointing to a person, Aman said, "His only brother is the father of my daughter's father". How is the person related to Aman?
A. Nephew B. Father C. Uncle D. Brother E. None of these

Q27. Pointing to a photograph a lady tells Manav, " I am the only daughter of this lady and her son is your maternal uncle." How is the speaker related to Manav's father?
A. Father B. Wife C. Uncle D. Son-in-law E. None of these

Q28. At what time between 5.30 and 6 will the hands of a clock be at right angles?
A. $43\frac{5}{13}$ min. past 5 B. $43\frac{7}{11}$ min. past 5 C. 40 min. past 4
D. 45 min. past 5 E. None of these

Q29. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through
A. 140^0 B. 145^0 C. 155^0 D. 175^0 E. None of these

Q30. How many times are the hands of a clock at right angle in a day?
A. 24 B. 34 C. 44 D. 54 E. None of these

Q31. If Rs. 400 becomes Rs. 1000 in 20 years at simple interest, the rate % p.a. is
A. 15 % B. 20 % C. 30 % D. 35 % E. None of these

Q32. Anoop borrowed Rs. 800 at 6 % p.a. & Rs. 1200 at 7 % p.a. for the same duration. He had to pay Rs. 1584 in all as interest. Find the time period.
A. 10 years B. 11 years C. 12 years D. 13 years
E. 15 years

Q33. Find out the capital required to earn a monthly interest of Rs. 210 at 7 % per annum simple interest.
A. Rs. 24000 B. Rs. 30000 C. Rs. 18000 D. Rs. 36000
E. None of these

Q34. Arun took a loan of Rs. 1400 with simple interest for as many years as the rate of interest. If he paid Rs.686 as interest at the end of the loan period, what was the rate of interest?
A. 8% B. 6% C. 4% D. 7% E. None of these

Q35. A sum of money at simple interest amounts to Rs. 850 in 3 years and to Rs. 930 in 4 years. The sum is :
A. Rs. 700 B. Rs. 690 C. Rs. 650 D. Rs. 610
E. None of these

Q36. Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?
A. Rs. 6400 B. Rs. 7200 C. Rs. 6500 D. Rs. 7500 E. None of these

Q37. There is 80% increase in an amount in 8 years at simple interest. What will be the compound interest of Rs. 14,000 after 3 years at the same rate?
A. Rs.3794 B. Rs.3714 C. Rs.4612 D. Rs.4634 E. None of these

Q38. The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re. 1. The sum is:
A. Rs.600 B. Rs.645 C. Rs.525 D. Rs.625 E. None of these

Q39. What is the difference between the compound interests on Rs. 5000 for 1.5 years at 4% per annum compounded yearly and half-yearly?
A. Rs. 2.04 B. Rs. 4.80 C. Rs. 3.06 D. Rs. 8.30 E. None of these

Q40. A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:
A. Rs. 120 B. Rs. 121 C. Rs. 123 D. Rs. 122 E. None of these

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Mid Term Examination, Odd Semester 2019-20

Program: B.Tech.(CSE) **Year: II** **Semester: III**

Sub Code: BCSC0010 **Sub Name: Discrete Mathematics**

Time: 1 Hour **Maximum Marks: 15**

Section – A

Note: Attempt All Questions. **3 x 2 = 6 Marks**

Q.1 Write the negation, converse, inverse and contrapositive of the statement: **"If it rains then the crop will grow."**

Q.2 (a) Given the truth values of p and q as T and those of r and s as F, find the truth value of the following:

$$(\neg (p \wedge q) \vee \neg r) \vee ((q \leftrightarrow \neg p) \rightarrow (r \vee \neg s))$$

(b) Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology without using truth table.

Q.3 Write the formula of the statement using predicate & quantifier and also obtain negation of the statement: **"Some students of Computer Engineering are good in mathematics."**

Section – B

Note: Attempt All Questions. **3 x 3 = 9 Marks**

Q.1 (a) Obtain the principal conjunctive normal form of the following

$$(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$$

(b) Show that $p \rightarrow (\neg q \vee r)$ is logically equivalent to $(p \wedge q) \rightarrow r$

Q.2 (a) Check the validity of the following arguments-

If it rains, then walking shall be troublesome. If my friend accompanies me, walking shall not be troublesome. My friend accompanies me. Therefore, it does not rain.

(b) Prove using mathematical induction that for all $n \geq 1$

$$1+4+7+...+(3n-2)=\frac{n(3n-1)}{2}$$

Q.3 A box containing 5000 IC chips, of which 1000 are manufactured by company A and the rest by company B. 10% of the chips made by company A and 5% of the chips made by company B are defective. If a randomly chosen chip is found to be defective, find the probability that it came from company A.

No of Pages:02

Second Mid-Term Examination, 2019-20
Course: B.Tech. (CSE), Year:II, Semester:III
Software Engineering (BCSC 0009)

Uni. Roll No: _____

Time: 1 Hrs. **Total Marks: 15**

Section A

Note: Attempt All Questions (3x2=6)

I. Explain the relationship between cost of effort and number of modules with the help of diagram.

II. Create flowchart for Test driven development process.

III. Consider a payroll management system to calculate payroll of employee. System can record daily attendance of employees and can generate summary of payroll of all employees on the basis of number of working hours, Gross pay and tax deduction.
Design the structured chart for the above system with proper use of Data flow, Control flow, Conditional call and Repititive call:

Section B

Note: Attempt All Questions (3x3=9)

I. Define the object types used to create Sequence diagram? Assume that you are working on a designing phase of a module related to Library Management System of an institution. The Requirement Analysis & Specification team has given you a case related to the project as: "Whenever a member asks for a book, the librarian checks the availability of book, if the book is available then the rack number of that book will be returned to librarian. Librarian then checks the validity of member by verifying the library card. If member is valid then the number of books issued to him and

transaction is created. Librarian then update the number of books issued to member and status of book."

Design a sequence diagram for the above case by using different types of objects.

II. Define module cohesion and explain any five different types of cohesion.

III. Consider the program given below. List out the operators, operands and Calculate the values of software science measures like Vocabulary, Length and Volume of the program.

```
int sort (int x[ ], int n)
{
    int i, j, save, im1;
    If (n< 2) return 1;
    for (i=2; i< =n; i++)
    {
        im1=i-1;
        for (j=1; j< =im1; j++)
            if (x[i] < x[j])
            {
                Save = x[i];
                x[i] = x[j];
                x[j] = save;
            }
    }
    return 0;
}
```

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Second Mid-Term Examination, Odd Semester 2019-20

Program: B.Tech (CCV), Year: II, Semester: III

Subject Name : Introduction to IT Infrastructure Landscape , Subject Code: BCSE0501

Time : 1 Hour **Maximum Marks: 15**

Section – A

Three questions of 02 marks each (with no internal choice). 3 x 2 = 6 Marks

1. "Firewall is nothing but a filter." Analyze the statement stating clearly the purpose of using a firewall. Also explain its types with example.

2. Discuss the types of network security zones found in an organization.

3. What is routing ? Explain the functions of a Router.

Section – B

Three questions of 03 marks each (with no internal choice). 3 x 3 = 9 Marks

1. "VLAN provides multiple benefits in cloud architecture." Comment on the statement clearly listing the benefits of using a VLAN. Also discuss the situations when VLANs are required.

2. "Topologies are either physical or logical." With reference to the statement discuss all network topologies with their advantages and disadvantages.

3. Explain how accountability is ensured using a digital signature with the help of a labeled diagram.

Printed Pages: 1 **University Roll No.....**

Mid-Term Examination, Odd Semester 2019-20

Program: B.Tech(CSF) Year:II, Semester:III

BCSE-0601 (IT Application & Data Security)

Time: 1 Hour **Maximum Marks: 15**

Section – A

3 x 2 = 6 Marks

Q.1 What do you understand by session management? Discuss and explain various session management vulnerability.

Q.2 Enlist the requirements of configuration management. Discuss the activities cover by configuration management.

Q.3 How Auditing and logging can be used for detecting the suspicious activity? Discuss the threats related to auditing and logging.

Section – B

3 x 3 = 9 Marks

Q.1 RSA system was used to encrypt the message M into the cipher-text C=6. The public key is given by n=p*q =187 and e = 107. In the following, we will try to crack the system and to determine the original message M.

a) Determine the private key for the given system.

b) What is the original message M?

Q.2 Use the Playfair cipher to decrypt the message "**KGBDKGYOBNCQFXHDTPMCNB**". The secret key can be made by filling the first row with the word **"TODAY"** and filling the rest of the matrix with the rest of the alphabet.

Q3 Write short note on any two:

a) Black box or system testing

b) White box or source testing

c) Stream cipher or block cipher

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Second Mid-Term Examination, Odd Semester 2019-20

B.Tech CSE (Data Analytics), II, III

BCSE 0551 : Introduction to Business Analytics

Time : 1 Hour **Maximum Marks: 15**

Section – A

Attempt all Questions. 3 x 2 = 6 Marks

I. Answer the following:

a. What is Business Intelligence?

b. What is a Point Solution?

II. Answer the following:

a. What is difference between supervised and unsupervised learning?

b. In what ways data mining is proved to useful in:

i. Sales

ii. Banking

III. What are the different data mining techniques? Explain any two with their applications.

Section – B

Attempt all questions. 3 x 3 = 9 Marks

I. What is data mining? What are the challenges one face in field of data mining?

II. What is CRISP-DM? In what sequence its phases run? It is the strict sequence?

III. Discuss BI Design. What are stages involving in its implementation with the breakdown of each stage in steps?

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Second Mid-Term Examination, Odd Semester 2019-20

B.Tech. (CS-IoT), Year - II, Sem. - III

BCSE 0651: Wireless Sensor Networks & IoT Standards

Time: 1 Hour **Maximum Marks: 15**

Section – A

Attempt **All** questions. 3 x 2 = 6 Marks

I. Explain the concept of Gossiping protocol.

II. What is meant by simulation? Name few simulators of wireless sensor networks and the language on which they are based.

III. Explain in detail why a routing protocol is required. Also mention advantage of hierarchical-routing protocols over position-based routing protocols.

Section – B

Attempt **All** questions. 3 x 3 = 9 Marks

I. Differentiate between the following:
   a. SPIN-PP and SPIN-EC
   b. SPIN-PP and SPIN-BC

II. Explain the advantages and disadvantages of Flooding protocol.

III. Differentiate between LEACH and PEGASIS in terms of cluster formation, cluster-head selection, energy efficiency and delay along with proper diagram of both.